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Printed Pages : 4

EAS401

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 3987

Roll No. | | | | | | | | | | |

B.Tech.

(SEMESTER-IV) THEORY EXAMINATION, 2011-12

MATHEMATICS – III

Time : 3 Hours] *[Total Marks : 100*

Note : Attempt questions from each Section as indicated. The symbols have their usual meaning. Provide statistical tables which are required by students.

Section – A

Attempt **all** parts of this question. Each part carries **2** marks : $10 \times 2 = 20$

1, (a) If $f(z) = u + iv$ is analytic, then show that the family of curves $u(x, y) = c_1$ and $v(x, y) = c_2$ are mutually orthogonal.

(b) Define removable and essential singular points with example.

(c) Define the coefficients of Skewness and Kurtosis.

(d) What is the total probability theorem ?

(e) Explain in brief Null and Alternative hypotheses.

(f) Define coefficient of contingency.

(g) Isolate the roots of the equation $x^3 - 4x + 1 = 0$.

(h) Verify that $\nabla E \equiv \Delta$.

(i) What do you mean by numerical differentiation ? Explain in brief.

(j) Let $I = \int_{x_0}^{x_3} f(x)dx$, where $f(x)$ is a third degree polynomial. Write the formula you will like to use to find the approximate value of I . It is given that the data are equispaced.

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Section – B

2. Attempt any **three** parts of this question. **3 × 10 = 30**

(a) Verify Cauchy's theorem by integrating exp (iz) along the boundary of the triangle with vertices at the points 1 + i, –1 + i and –1 – i.

(b) Find all four central moments and dicuss skewness and kurtosis and also Karl Pearson skewness for the frequency distribution given in the following table :

Range of expenditure (in ₹ 100 per month)	2 – 4	4 – 6	6 – 8	8 – 10	10 – 12
No. of families	38	292	389	212	69

(c) A manufacturer claimed that at least 95% of the equipments which he supplied to a factory conformed to the specifications. An examination of a sample of 200 pieces of equipments revealed that 18 were faulty. Test this claim at a significant level of (i) 0.05 and (ii) 0.01.

(d) Show that the Newton-Raphson Method has second order convergence.

(e) Solve the following system using Crout's decomposition method :

$3x - y + 2z = 12$

$x + 2y + 3z = 11$

$2x - 2y - z = 2$

Section – C

All questions of this section are compulsory. Attempt any **two** parts from each question. **5 × 10 = 50**

3. (a) Determine p such that the function

$$f(z) = \frac{1}{2}\log(x^2 + y^2) + i\tan^{-1}\left(\frac{px}{y}\right)$$ is an analytic function. Also find f'(z).

(b) Find Laurent series expansion of

$$f(z) = \frac{4z - 1}{z^4 - 1}$$

about the point z = 0.

(c) Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5 + 4\cos\theta}\, d\theta$.

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4. (a) Fit a parabola of the form $y = a + bx + cx^2$ to the data

x	1	2	3	4
y	1.7	1.8	2.3	3.2

by the method of least squares.

(b) If θ is the acute angle between the two regression lines in case of two variables x and y, show that

$$\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2},$$

where r, σ_x, σ_y have their usual meanings. Explain the significance of the formula when $r = 0$ and $r = \pm 1$.

(c) Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys ? Assume equal probabilities for boys and girls.

5. (a) Fit a binomial distribution to the data given in the following table :

x	0	1	2	3	4
f	24	41	28	5	2

(b) The number of scooter accidents per month during a year in a certain town were as follows :

12, 8, 20, 2, 14, 10, 15, 6, 9, 4, 7, 13

Are these frequencies in agreement with the belief that the accident conditions were the same during the whole year ?

(c) $\overline{X}$ and R values for 10 sub-groups of 5 readings are given in the following table.

Determine the control limits for $\overline{X}$ and R charts for future use, eliminating all the out of control points :

Sub-group numbers	1	2	3	4	5	6	7
$\overline{X}$	34.0	31.6	30.8	33.0	35.0	32.2	33.0
R	4	4	2	3	5	2	5

Sub-group numbers	8	9	10
$\overline{X}$	32.6	33.8	37.8
R	13	19	6

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(12)

6. (a) Find the root of $2 \sin x - 2x + 1 = 0$ correct to five significant digits with initial approximation $x_0 = 1.0$.

(b) Estimate the values of a and b in the following table :

X	10	15	20	25	30	35
$f(x)$	43	a	29	32	b	77

(c) The population of a town in decennial census is as under. Estimate the population for the year 1955 :

Year	1921	1931	1941	1951	1961
Population (in lac)	46	66	81	93	101

7. (a) Find $\frac{d}{dx}(J_0)$ at $x = 0.1$ from the following table :

x	0.0	0.1	0.2	0.3	0.4
$J_0(x)$	1.0000	0.9975	0.9900	0.9776	0.9604

Also find $\frac{d^2}{dx^2}(J_0)$ at $x = 0.1$.

(b) Evaluate the integral

$$\int_0^{2\pi} e^{-t} \sin(10\,t)\, dt$$

using Simpson's 3/8 rule.

(c) Apply Picard's method to find the solution of the initial value problem

$$\frac{dy}{dx} = y - x,\ y(0) = 2$$

Show that the iterative solution approaches the exact solution.

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